

Bachelor of Technology (Computer Science & Engineering)

CSD493: Project-1, Monsoon 2026

Project Proposal Form

Date: 23rd May 2026

Project Title: Behavioral-Diff Detection of Malicious Browser Extension Updates

Area of Specialization: NA

Names and SNU IDs of Group Members (in capital letters):

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Name of Project Advisor: Dr. Sweta Mishra

Project Proposal Description:

MOTIVATION:

Browser extensions update silently and automatically. A trusted extension can become malicious overnight - the December 2024 Cyberhaven supply chain attack pushed a credential-stealing payload to 400,000 active users within hours, entirely undetected. Static code review and permission analysis fail against obfuscated payloads and behaviors that only surface at runtime under specific browsing conditions. There is no open, systematic tool that detects malicious changes between extension versions through dynamic behavioral comparison.

PROBLEM STATEMENT:

This project builds a sandboxed behavioral-diff framework that flags malicious browser extension updates by comparing runtime traces across consecutive versions. The system executes both version N and version N+1 inside an isolated Docker container under identical Playwright-automated browsing tasks, recording outbound network calls, DOM field access, Chrome API usage, cookie reads, and tab interactions for each version. A weighted feature-difference vector is computed between both traces and scored by a Random Forest classifier trained on confirmed malicious update pairs from the 2024 supply chain incidents and synthetically weaponized open-source extensions. The output is an explainable per-update diff report that specifies exactly which new behaviors triggered the alert - giving security teams actionable intelligence rather than opaque risk scores. The system will be evaluated on detection rate, false positive rate, and per-update latency overhead.

Signature of Group Member(s), Date	Signature of Project Advisor, Date